
stripe + VISA

HISTORY



Stripe

Stripe

Stripe is a financial technology company that provides online payment services for businesses and e-commerce websites.

Found (2010)

Stripe was founded by **Patrick Collison and John Collison**, two brothers from Ireland, in 2010.

Process

They created Stripe because accepting payments online was difficult for developers and businesses. Their goal was to make online payments easier to integrate using software and APIs.

HISTORY



Visa

Visa

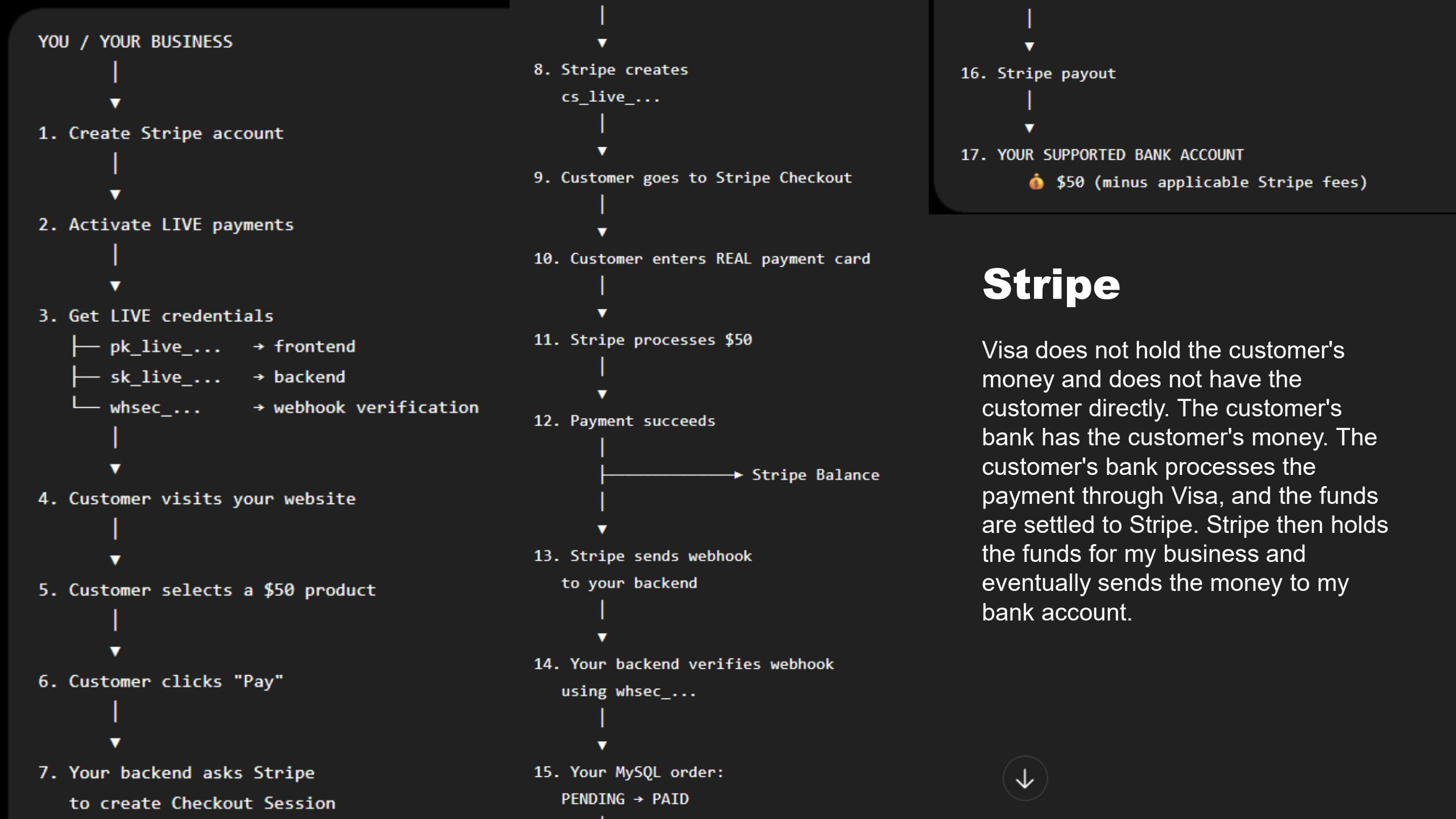
Visa is a global financial technology company that operates one of the world's largest electronic payment networks. It helps people and businesses make payments using debit cards, credit cards, and other electronic payment methods.

Beginning (1958)

Visa's history began in **1958**, when Bank of America launched the **BankAmericard** credit card in California, USA.
It was one of the first general-purpose credit cards that allowed customers to borrow money for purchases.

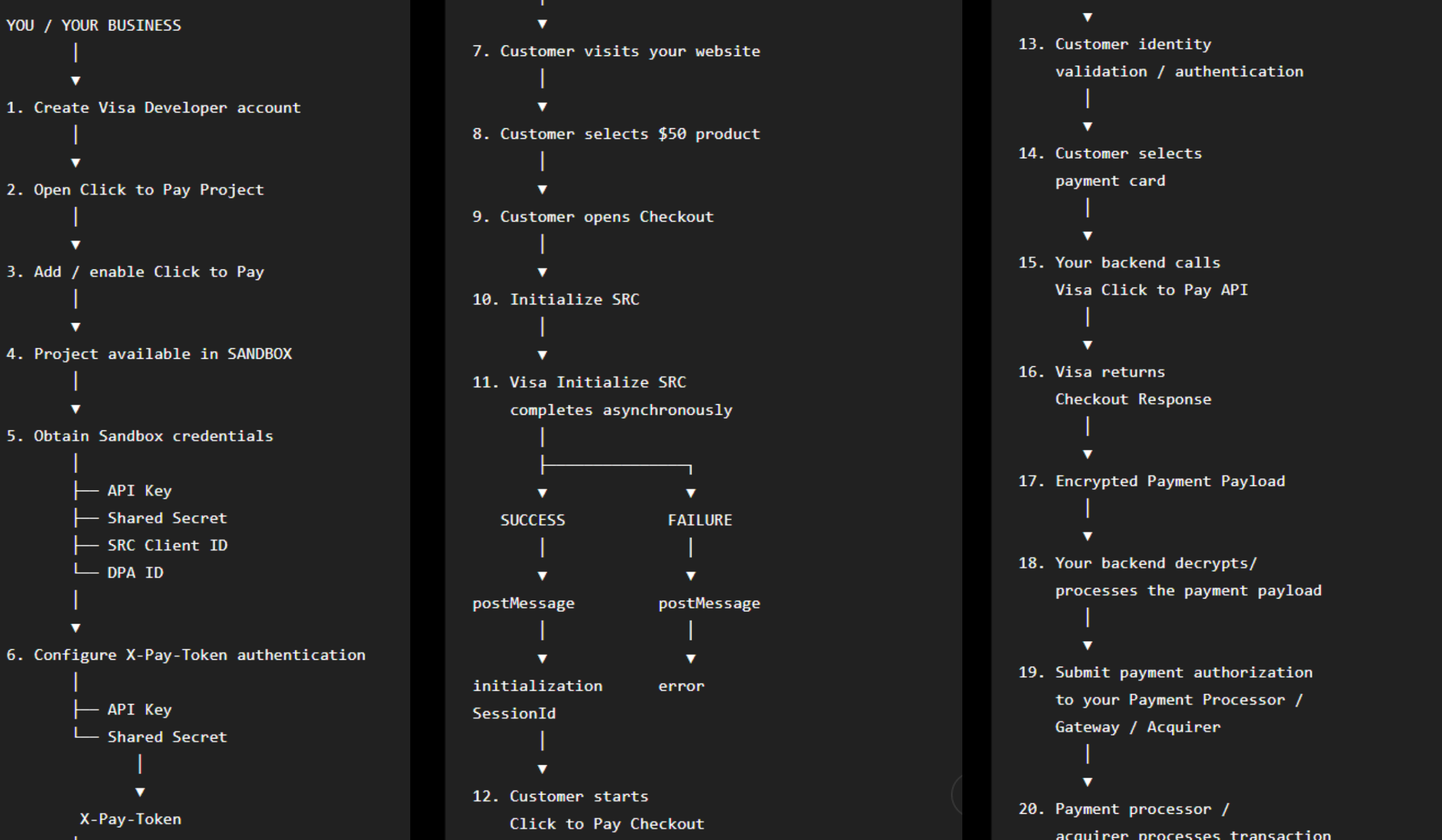
Process

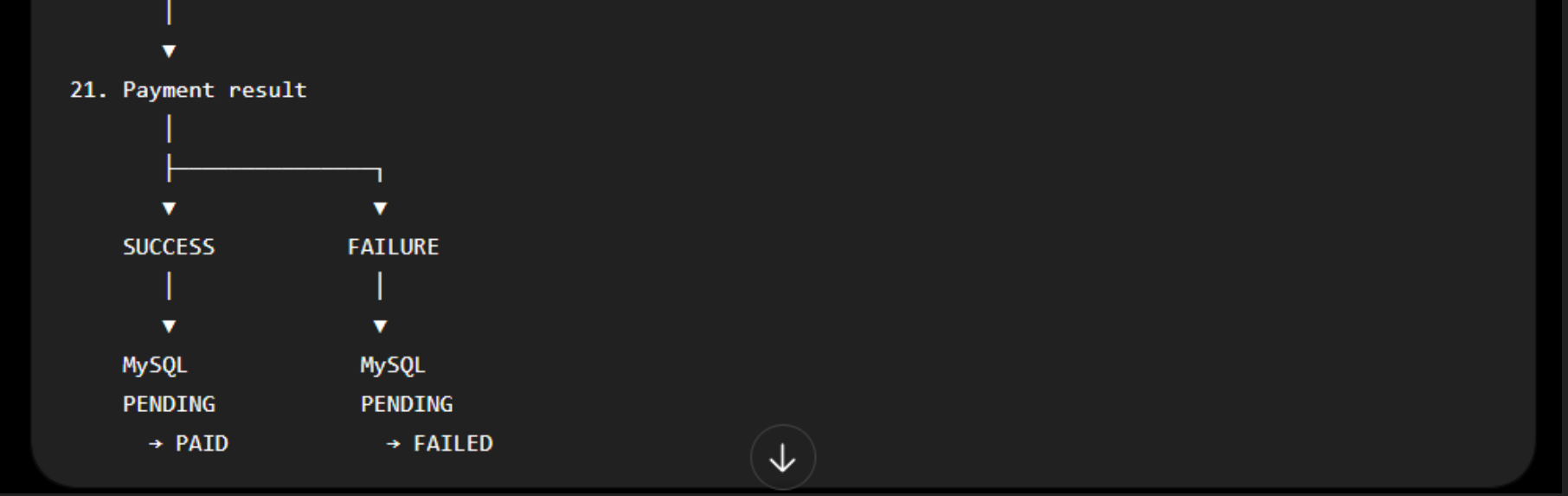
Visa **does not generally lend money directly to cardholders**. Instead, it provides the payment network that allows banks and businesses to process transactions.



Stripe

Visa does not hold the customer's money and does not have the customer directly. The customer's bank has the customer's money. The customer's bank processes the payment through Visa, and the funds are settled to Stripe. Stripe then holds the funds for my business and eventually sends the money to my bank account.


YOU / YOUR BUSINESS
1. Create Visa Developer account
2. Open Click to Pay Project
3. Add / enable Click to Pay
4. Project available in SANDBOX
5. Obtain Sandbox credentials
API Key
Shared Secret
SRC Client ID
DPA ID
6. Configure X-Pay-Token authentication
API Key
Shared Secret
X-Pay-Token
7. Customer visits your website
8. Customer selects $50 product
9. Customer opens Checkout
10. Initialize SRC
11. Visa Initialize SRC completes asynchronously
SUCCESS
FAILURE
postMessage
postMessage
initialization SessionId
error
12. Customer starts Click to Pay Checkout
13. Customer identity validation / authentication
14. Customer selects payment card
15. Your backend calls Visa Click to Pay API
16. Visa returns Checkout Response
17. Encrypted Payment Payload
18. Your backend decrypts/ processes the payment payload
19. Submit payment authorization to your Payment Processor / Gateway / Acquirer
20. Payment processor / acquirer processes transaction



VISA

Visa does not hold the customer's money and does not have the customer directly. The customer's bank (issuing bank) has the customer's money. The customer's bank processes the payment through Visa, and the funds are settled to **your acquiring bank (acquirer)**. **The acquirer holds/settles the funds for your business** and eventually transfers the money to your bank account.

Stripe

Stripe item	What it is for
Sandbox	Used to **test payments**. No real money is charged.
Stripe Live	Used to **accept real payments** from real customers. Real money can be transferred.
Publishable Key [pk_...]	Used by the **frontend** to identify your Stripe account when using Stripe.js or other client-side Stripe features.
Secret Key [sk_...]	Used by your **backend** to securely communicate with Stripe's API, such as creating Checkout Sessions. 🔒
Webhook Signing Secret [whsec_...]	Used by your **backend** to verify that a webhook request actually came from Stripe. 🔒
Checkout Session ID [cs_]	Identifies **one specific Stripe Checkout session**. Stripe gives this to your application after you create the session.

VISA

Stripe item	What it is for
API Key	Visa-provided identifier for your application/project. It is sent with applicable API requests and, for Click to Pay, is one of the authentication mechanisms.
Shared Secret	The **decrypted Visa shared secret**. It is sensitive and is used with the API key to generate the **X-Pay Token** for APIs that require it. .
Public Key	The matching **public key**. It can be shared with Visa when the integration uses public/private-key authentication or encryption.
Private Key	Your **private cryptographic key**. Used for operations such as signing/authentication when that Visa integration requires asymmetric keys. **Never expose it to the frontend.**
Encrypred Shared Secret	The **encrypted form of the Visa shared secret**. It protects the shared secret while stored.
Decoded Data	The **decoded/decrypted binary data** produced while processing the encrypted secret. It's an intermediate file, not normally something your frontend needs.

Stripe Dashboard vs Stripe CLI

Neither is "better" overall because they serve two completely different purposes; **the Dashboard is best for configuration, viewing data, and production**, while **the Stripe CLI is best for local coding, testing, and debugging.**

Feature	Stripe Dashboard	Stripe CLI
Best For	Business management, analytics, monitoring production webhooks, manual refunds, and creating products.	Writing code, testing localhost webhooks, triggering mock events, and automating developer workflows.
Interface	Web UI (visual, click-based).	Command Line / Terminal.
Localhost Webhooks	Cannot reach localhost directly (requires a public tunnel like ngrok).	Routes directly to localhost automatically (stripe listen).
Testing Events	Requires manual checkout UI actions or re-sendi ng past events.	Instant mock events with one command (e.g., stripe trigger payment_intent.succeeded).
Production Setup	Required. This is where you configure your live se rver's webhook URL.	Not used in production.

Stripe used

At-a-glance

Stripe is a payment processing platform that helps businesses accept payments from customers online or in person.

ACCEPT PAYMENTS

Customers can pay using Visa, Mastercard, and other payment methods.

20XX

Jan



ONLINE CHECKOUT

Connect your website to Stripe so customers can pay for products.

20XX

Feb



MANAGE PAYMENTS

Track successful payments, failed payments, and refunds.

20XX

Mar



RECEIVE MONEY

Transfer your business's payment funds to your bank account.

20XX

Apr



SUBSCRIPTIONS

Automatically charge customers regularly, such as monthly.

20XX

May



VISA used

At-a-glance

Visa is a global payment network that helps transfer payment information between customers, banks, and businesses.

PROCESS CARD PAYMENTS

Helps customers pay using Visa cards.

20XX

Jan

CONNECT BANKS AND BUSINESSES

Transfers payment information between the customer's bank and the merchant's bank.

20XX

Feb

AUTHORIZE TRANSACTIONS

Helps check whether a payment can be approved.

20XX

Mar

SUPPORT INTERNATIONAL PAYMENTS

Allows Visa cards to be used in many countries.

20XX

Apr

PROVIDE PAYMENT SECURITY

Helps detect and prevent fraudulent transactions.

20XX

May

How Stripe earns money

Stripe mainly earns money by charging **businesses/merchants fees for processing payments**.

The **actual Stripe fee depends on the country, payment method, and Stripe pricing**, so $1.75 is only an example.



Customer pays	$50.00	
Stripe fee	-1.75	2.9%

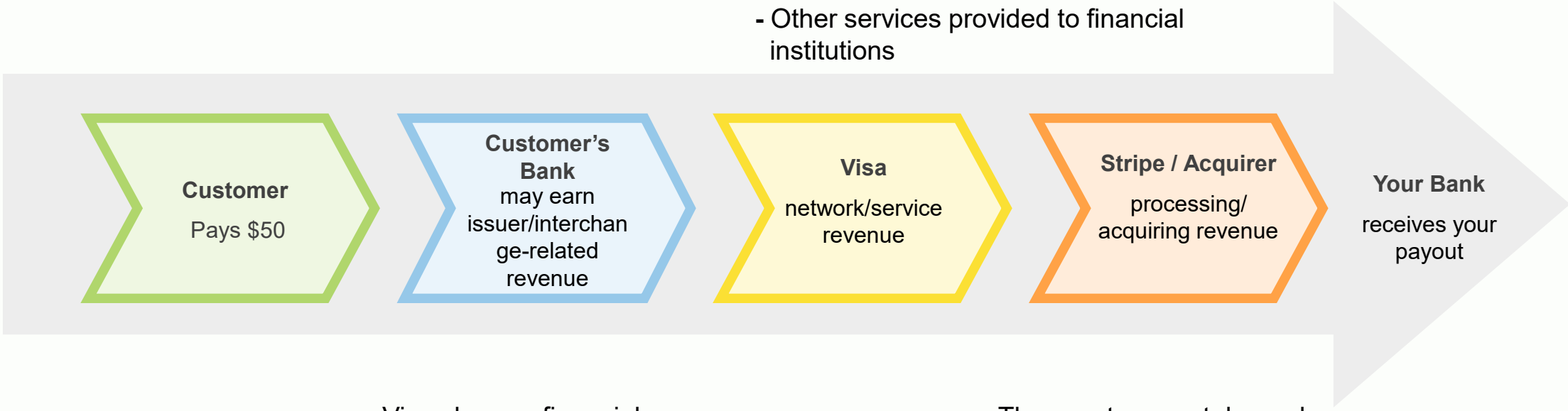
You receive	$48.25	

Stripe also makes money from other business services, such as billing, subscriptions, fraud tools, issuing, and other financial infrastructure.

How VISA earns money

Visa generally **doesn't make money by taking your customer's $50 and keeping part of it**.

Visa can earn fees from things such as:
- **Network/service fees**
- **Data processing fees**
- **International transaction-related fees**
- Other services provided to financial institutions



Visa charges financial institutions for using its payment network and related services.

The exact amount depends on the transaction and agreements between the parties.

Summary

VISA vs. Stripe

Visa: A payment network that connects the customer's bank and the merchant's bank. It helps process card transactions and earns network fees.

Stripe: A payment processing platform that helps your website accept payments. It charges processing fees and sends your money to your bank.

$50 Payment Flow

Stripe

Customer > Customer's Bank > Visa Network > Stripe > Your Stripe Balance > Your Bank Account

VISA

Customer > Customer's Bank > Visa Network > Your Bank Account

Simple summary: Visa connects the banks; Stripe processes your website payments and pays you.



END



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